The Field Equations of the Screened Khronon Action, and a Verdict for Every Physics Class

Working note (PAPER70), draft v0.1, 2026-11-07. AI-assisted research programme; not peer reviewed. Not deposited. Companions: PAPER67 (DOI 10.5281/zenodo.23968171, the law and its failure record), PAPER68 (DOI 10.5281/zenodo.23990944, the action) and PAPER69 (DOI 10.5281/zenodo.24025170, the closure record). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework.

Abstract

Gravity carries one extra scale, $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$, $\kappa = \frac{1}{2}$ fitted (proved not derivable), realised by the v3s screened khronon action. It passes Cassini, PPN, the gravitational-wave speed, galaxy lensing and SPARC, and fails AGC 114905, Malin 1 and the canonical Υ-tilt row. It also under-fills X-COP clusters, groups and Coma ultra-diffuse galaxies by amounts that vary about 100× between classes, after the IMF and tides account for the ellipticals, most of KiDS and the bright M31 dwarfs. Separately, it fails the CMB-lensing shape, which needs cold extra gravity at $z \simeq 3400$. This note writes down the action in the Legendre form every lane solved (correcting the form printed in PAPER68), its complete covariant field equations checked against direct variation, their 3+1 form and a convergent static solver. It then lists every gate with its frozen label. Four physical classes of extra ingredient were built and graded on both footings: a cold fluid, a warm fixed-temperature sea, a hot thermal fermion, and a shock-entropy sector written into the action. Each meets its own wall, and no candidate passes every gate. The most probable reading of the record is the deposited, scoped law plus separate per-class systematics, with the cluster, group and ultra-diffuse rows and the CMB wall left unexplained.

1 Conventions

Units $c = 1$ and $K = 4\pi G$ unless stated; restoring c: $K \to 4\pi G/c^2$, the prefactor of R is $c^4/16\pi G$, and $a_0 \to a_0/c^2$ inside W. Two footings are carried and never pooled: canonical $a_0 = 9.3603 \times 10^{-11}$ and alt $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. κ is fitted; a Lean proof shows that it is not derivable from the programme's premises. A gate is closed when it carries a label fixed by criteria committed before the first graded number; every lane runs a mutation twin that must fail.

2 The action, in Legendre form

PAPER68 prints $S = \frac{1}{4K}\int\sqrt{-g}\,[R - 2\mathcal{J}(Y) + 2\mathcal{K}(Q)] - \frac{\gamma_L}{2}\int\sqrt{-g}\,[\max(D\cdot P, 0)]^2 + S_m$. Read literally, P then appears only in the stiffness, its own equation forces $D\cdot P$ to be constant, and P decouples from gravity. The form every lane has actually solved puts $\mathcal{J}$ in Legendre form (QX17):

$$\frac{\mathcal{L}}{\sqrt{-g}} = \frac{R - 2\Lambda + 2\mu_K^2(Q-1)^2}{4K} + P^m A_m - W(K|P|) - \gamma_L F(\Pi) + \lambda\, n^m P_m, \qquad W(p) = \frac{a_0^2}{K}\Big[-\frac{p}{a_0} - \ln\Big(1 - \frac{p}{a_0}\Big)\Big], \tag{1}$$

with $F(\Pi) = \frac{1}{2}[\max(\Pi, 0)]^2$, $\Pi = D_m P^m$ and $\gamma_L = KL_\star^2/2$, plus minimally coupled matter S_m. Maximising over P returns the printed $\mathcal{J}$ exactly: $\max_P[P\cdot A - W(K|P|)] = (a_0^2/K)[u - \ln(1+u)]$ with $u = |A|/a_0$, attained at $K|P| = a_0 u/(1+u)$ (sympy, residual 0). So at $\gamma_L = 0$, Eq. (1) is the printed action; with $\gamma_L > 0$, P is an independent field and Eq. (1) is the theory. **This is an erratum on PAPER68's printed form only; no number changes.**

Definitions. τ is the khronon; $Q = \sqrt{-g^{mn}\partial_m\tau\,\partial_n\tau}$ (= $1/N$ in unitary gauge); $n_m = -\partial_m\tau/Q$; $h_{mn} = g_{mn} + n_m n_n$; $l = \ln Q$; $A_m = n^n\nabla_n n_m = -h_m{}^n\partial_n l$; $D_m = h_m{}^n\partial_n$; $\Pi = \nabla_m P^m - P^m A_m$; $\Pi_+ = \max(\Pi, 0)$; $\hat{P} = P/|P|$; $\mathcal{D}_m = \partial_m\Pi_+ - \Pi_+\partial_m l$.

3 The field equations (QX17, CERTIFIED)

$$\text{(E1)}\quad n^m P_m = 0, \tag{2}$$

$$\text{(E2)}\quad (1+\gamma_L\Pi_+)A_m + \gamma_L D_m\Pi_+ = KW'(K|P|)\,\hat{P}_m, \qquad \lambda = (1+\gamma_L\Pi_+)\,n\cdot\partial \ln N + \gamma_L\, n\cdot\partial\Pi_+, \tag{3}$$

$$\text{(E3)}\quad \nabla_m\big[n^m\rho_Q - \lambda P^m/Q\big] = 0, \qquad \rho_Q = \frac{\nabla_m[(1+\gamma_L\Pi_+)P^m]}{Q} + \frac{\mu_K^2(Q-1)}{K}, \tag{4}$$

$$\text{(E4)}\quad G_{mn} + \Lambda g_{mn} = 2K\big(T^{\rm mat}_{mn} + T^{Q}_{mn} + T^{W}_{mn} + T^{P}_{mn}\big), \tag{5}$$

with

$$T^{Q}_{mn} = g_{mn}\frac{\mu_K^2(Q-1)^2}{2K} + \frac{\mu_K^2}{K}Q(Q-1)\,n_m n_n, \qquad T^{W}_{mn} = -g_{mn}W + KW'\,\frac{P_mP_n}{|P|},$$

$$T^{P}_{mn} = g_{mn}(P\cdot A) + 2P_{(m}\partial_{n)}l + (\nabla\cdot P)\,n_m n_n - 2\lambda\, n_{(m}P_{n)} + \gamma_L\big[-g_{mn}F + 2Fh_{mn} + h_{mn}P^r\mathcal{D}_r - 2P_{(m}\mathcal{D}_{n)}\big].$$

(E5) Matter is minimally coupled, $\nabla^m T^{\rm mat}_{mn} = 0$. The energy density seen by the foliation is $n^m n^n T_{mn} = (\mu_K^2/2K)(Q^2-1) + W + \Pi + \gamma_L F$. At $\gamma_L = 0$ the leaf part of (E2) inverts to $K|P| = a_0|E|/(a_0+|E|)$, the Legendre map; Π_+ is C^1, so the equations hold in the weak sense across the switch surface.

Checks. Each equation is compared with a direct Euler–Lagrange variation of Eq. (1) on jet symbols (mpmath, 40 digits, three random configurations in 2+1 dimensions): P equation max rel 3.8×10^{-41}, khronon equation 1.8×10^{-41}, full stress tensor 1.2×10^{-40}; FRW reduction exact. The MUTATE twin (two flipped signs) fails at 0.17–0.78.

AC2q (candidate, not adopted; QX17A, CERTIFIED). The dark-energy scalar is the clock, there is no $\mathcal{K}(Q)$ term, and the switch acts on the khronon charge $\Sigma = \Pi + Q^2$: $\mathcal{L}/\sqrt{-g} = R/4K + X - V(\phi) + P\cdot A - W - \gamma\{F(\Pi+2X) - (2X)^2/2\} + \lambda n\cdot P$, $X = Q^2/2$. Its P equation is (E2) with $\Pi_+ \to \Sigma_+$; its clock equation is $\nabla_m[n^m\rho_\phi - \lambda P^m/Q] = -V'(\phi)$ with $\rho_\phi = J_P/Q + Q[1-2\gamma(\Sigma_+ - Q^2)]$, which is $\Box\phi = V'$ at $P=\gamma=0$; at $P=0$ the stiffness stress vanishes identically, so the background is the canonical scalar's. The MUTATE twin fails at $0.23/0.39/7.3\times10^{-3}$.

4 3+1 form, reductions and solutions

ADM (QX17B). P and N carry no time derivatives, so the P equation and the lapse equation are constraints on each leaf. The P sector's momentum density vanishes on shell, so the momentum constraint is GR's; the tensor principal part is GR's, so $c_T = c$. The Lean kernel file for these steps has 12 theorems, standard axioms only, and its MUTATE twin fails. The lane as a whole is not certified: its numerical ADM row failed as frozen on a sign convention in the checker's off-shell momentum density, and its `solve_bvp` spherical solver did not converge.

Reductions. FRW with $P=0$: $\bar{Q} = 1 + I_0/(2\mu_K^2 a^3)$ and a dust-like $\rho \propto a^{-3}(1 + w_0/a^3)$. Static, weak, $\gamma_L = 0$: $\nabla\cdot[\mu(|\nabla\Phi|/a_0)\nabla\Phi] + \mu_K^2\Phi = 4\pi G\rho_b$, $\phi = \psi$. Spherical, $\mu_K = 0$: the law $g_N = g^2/(g+a_0)$ (Lean). Static, $\gamma_L > 0$: the Cassini hedgehog inside $L_\star$.

Solutions (QX17C, CERTIFIED as a solver). A conservative finite-difference Newton solver with a decaying Bessel outer boundary converges in 1–6 steps on both footings, about 5 s in all. It reproduces the law at $L_\star \to 0$ (worst 3.9×10^{-7}), the solar hedgehog to 7.1×10^{-6} of $g_\star = GM/L_\star^2$, and a $10^{10}M_\odot$ galaxy on the law to 4.8×10^{-6}; full-GR minus weak-field is 4.1×10^{-6} (Sun) and 5.4×10^{-7} (galaxy). An isolated point Sun reaches only 0.71/0.67 of the law at $10^3 L_\star$, so whether a galaxy built from screened stars shows the law is open.

5 Every gate, with its frozen label

PAPER69's verdict table is carried unchanged. The rows below are the 2026-11-06 lanes that came after it.

Gate	Label	Lane
Equations and health		

Gate	Label	Lane
Covariant field equations, v3s and AC2q	CERTIFIED vs direct variation	QX17, QX17A
3+1 form, Lean kernels	Lean PASS; lane NOT CERTIFIED (numerical row, solver)	QX17B
Static spherical solver	CERTIFIED, both footings	QX17C
Π^6 repair, curved 3D spherical switch	PASS, not adopted; non-radial modes untested	BL3
AC2q decelerating slow roots	HARMLESS, scoped; $\epsilon_\phi = 2$ NOT REACHED by any record history	EP1, EP2
Nonlinear fold, shock-capturing	NO VERDICT as frozen (viscosity check fails); no pair where AC2q regularizes	FS1
Moving Sun convergence	NO VERDICT, bracketed; largest time-mean on any rung 0.147 of the line	MS1, MS2
Light-cone causality	FAIL (physical, far below data); no repair in the khronometric class	AC18, SV2
All-gates battery (static gates G1–G8; G9 CMB)		
Six candidates on G1–G8	NOTHING PASSES ALL STATIC GATES; every candidate fails the Coma UDGs by at least $3.97\,\sigma_U$	GB1
Cold fluid: $\mathcal{K}(Q)$ dust, amplitude freed	FAIL: passes the background and CMB lensing, fails G1–G7; cosmic capture 4.111 in every class	KD1
Cold component decaying late	FAIL: CMB-lensing χ^2 46.3/53.0 against 18.48 at the longest lifetime discs allow	LD1
Hot thermal fermion, 11.1377 eV	FAILS 7 of 9 gates; CMB-lensing χ^2 166.0–194.2	HC1
Shock-entropy field (proxy)	FAIL: X-COP and groups; need falls outward while the measured adiabat rises	EN1
Shock-entropy sector in the action	FAIL by the no-tuning rule; static solutions isentropic (f $\leq$ 0.04)	SE1
The step as several causes		
Multi-cause step	MULTI-CAUSE IN PART: IMF explains the ellipticals and KiDS	ST1
Joint systematics budget per class	IN PART: ellipticals, KiDS and bright M31 dSphs EXPLAINED; X-COP +0.440, groups +0.232, Coma UDGs +1.120 NOT EXPLAINED (canonical medians); CMB SEPARATE	SE2
Environment, re-read	NO: no environment key passes; satellites need about 1.6× the phantom, Coma UDGs 4–46×	EV1
Disc-only exclusion	every direction DEAD; even a perfect (oracle) exclusion fails GB1	DX1

6 What an all-gates ingredient would have to supply

On top of the law, the extra gravitating mass as a fraction f of the baryons (KD1 diagnostic, canonical / alt):

Where	Needed f
SPARC discs	≤ 0.33 / 0.19
X-COP clusters	~ 4.2 / 3.8
groups	1.2–3.5 / 0.9–2.9
X-ray ellipticals	~ 0.9 / 0.8 (about half of it IMF)
dSph satellites	~ 1.7 / 1.3
Coma UDGs	~ 80
the CMB	cold at z 3400 ($w \leq 0.01$), lensing $\chi^2 \leq 18.5$ at $L \leq 400$

7 Four walls

1. **Cold fluid** (KD1, QX3, QX7). Every well captures it, discs included. The CMB and BBN need $\mu_K \geq 8.9 \times 10^6\,\mathrm{Mpc}^{-1}$; galaxies allow at most 79.

2. **Warm fixed-temperature sea** (QX12). Retention $\exp(\text{depth}/c_s^2)$ is too sharp; the temperature would have to track each host.
3. **Hot thermal fermion** (HC1, HF1, VW1). Hot enough to stay out of discs, it under-fills clusters, cannot fill dSphs or UDGs, and fails CMB lensing; cold enough for the CMB, discs capture it. No window.
4. **Shock-entropy sector** (SE1). A conserved, non-cooling sector heated by its own shocks either sits isentropic, and holds almost nothing, or needs a different entropy history for each class at the same entropy. Its conserved rest mass also caps a host near the cosmic share, against the ~ 80 the Coma UDGs need.

The common reason: an ingredient that tracks each host's virial state has to virialise with the host; if it is also cold early, nothing keeps it out of discs. And the CMB needs the extra gravity at $z \simeq 3400$, where nothing is virialised, so any key that is on only in today's hot systems is off exactly where the CMB needs it on.

8 Reading

Every physics class on the record now carries a label, and none passes every gate. The law, and the action within its scope, describe cold rotating discs and the Solar System. The ellipticals, KiDS and the bright M31 dwarfs are accounted for by measured systematics. X-COP, the groups and the Coma UDGs are not, and the CMB-lensing wall stands on its own. This is the best remaining answer the record supports, not a closed theory.

9 What waits on the owner

The PAPER68 erratum for the Legendre form; adoption of the Π^6 stiffness repair; the deposit of this note.

Method

Every lane froze its criteria in a commit before any script ran, and every lane has a MUTATE twin that must fail. Footings are reported apart. No ΛCDM inputs were used. Scripts and outputs are in the repository under closure_2026_11_06/ at the commits the lane RESULT files name.